

16. Answer: C

Intensity of the wave is proportional to the square of the amplitude of the wave.

$$\text{Intensity of wave Y, } I_Y = k(A)^2$$

$$\begin{aligned}\text{Intensity of wave X, } I_X &= k(2A)^2 \\ &= 4kA^2 \\ &= 4 I_Y\end{aligned}$$

$$\text{Wavelength of wave Y, } \lambda_Y = \frac{v}{f}$$

$$\begin{aligned}\text{Wavelength of wave X, } \lambda_X &= \frac{v}{2f} \\ &= \frac{1}{2} \lambda_Y\end{aligned}$$

17. Answer: D